

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
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CHEMISTRY

9729/01

Paper 1 Multiple Choice

22 September 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

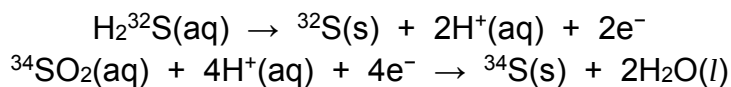
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 Equal volumes of 1 mol dm^{-3} hydrogen sulfide and sulfur dioxide (each containing a different isotope of sulfur) are mixed to precipitate sulfur according to the equations shown below:



What is the relative atomic mass of the sulfur precipitated?

- A** 32.1
B 32.7
C 33.0
D 33.3
- 2 50 cm^3 of a $0.0400 \text{ mol dm}^{-3}$ solution of acidified Fe^{2+} was oxidised to Fe^{3+} by 100 cm^3 of $0.0100 \text{ mol dm}^{-3}$ of an oxidising agent, XO_3^- .

What is the final product of **X**?

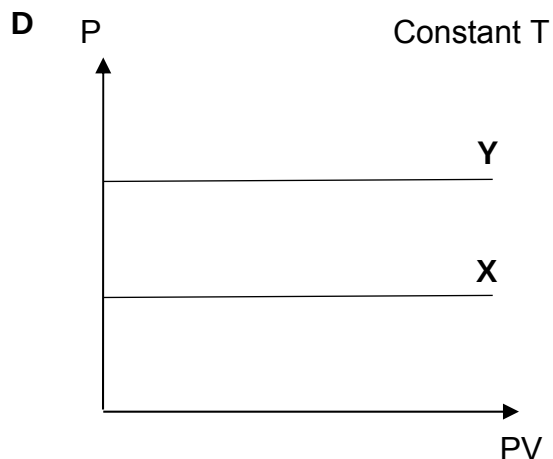
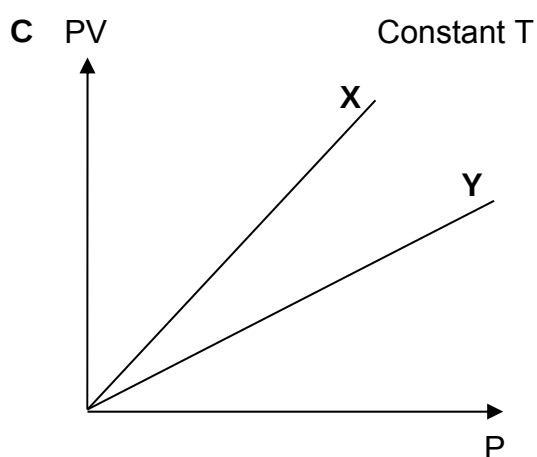
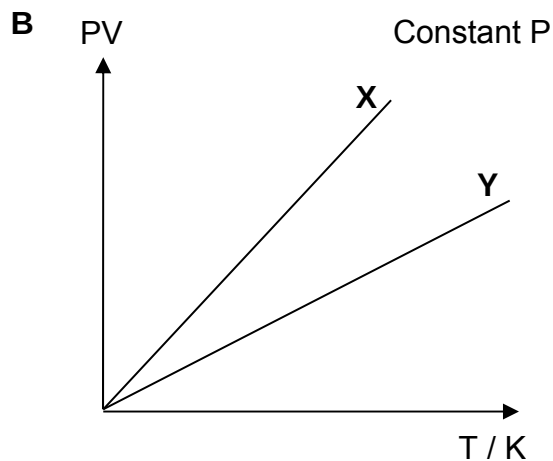
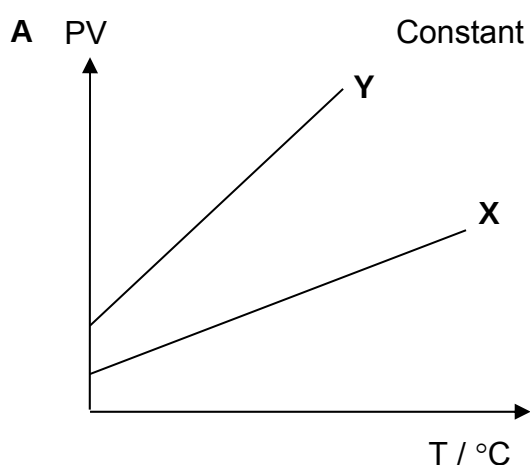
- A** XO_2
B XO^-
C XO_2^-
D X^-
- 3 Paramagnetism refers to the magnetic state of an atom with one or more unpaired electrons. The greater the number of unpaired electrons in an atom, the greater the paramagnetism.
- Which atom has the greatest paramagnetism?
- A** Oxygen
B Chlorine
C Scandium
D Vanadium

4 Which of the following statements about the nitrate(V) ion, NO_3^- , are correct?

- 1 The hybridisation of the nitrogen atom is sp^2 .
- 2 All the nitrogen–oxygen bond lengths in NO_3^- are equal in length.
- 3 NO_3^- has a total of twenty-four valence electrons.

- A 1, 2 and 3
 B 1 and 2 only
 C 1 and 3 only
 D 2 and 3 only

5 Which graph correctly describes the behaviour of fixed masses of the ideal gases X and Y, where X has a higher M_r than Y?



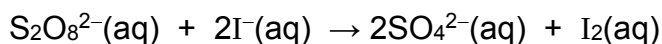
- 6 A student dissolved 10.5 g of sodium fluoride, NaF, in 250 g of water. The following data is obtained:

Lattice energy of NaF	-918 kJ mol ⁻¹
Enthalpy change of hydration of F ⁻	-457 kJ mol ⁻¹
Enthalpy change of hydration of Na ⁺	-390 kJ mol ⁻¹

Given that the specific heat capacity of water is 4.2 J g⁻¹ K⁻¹, what would be the final temperature of the solution, if the initial temperature of water is 30°C?

- A** 6.5°C **B** 13.1°C **C** 16.9°C **D** 46.9°C

- 7 The kinetics of the reaction between iodide and peroxodisulfate can be investigated by varying the volume of the reactants used. The two reactants are mixed in the presence of a known amount of Na₂S₂O₃ and a little starch.



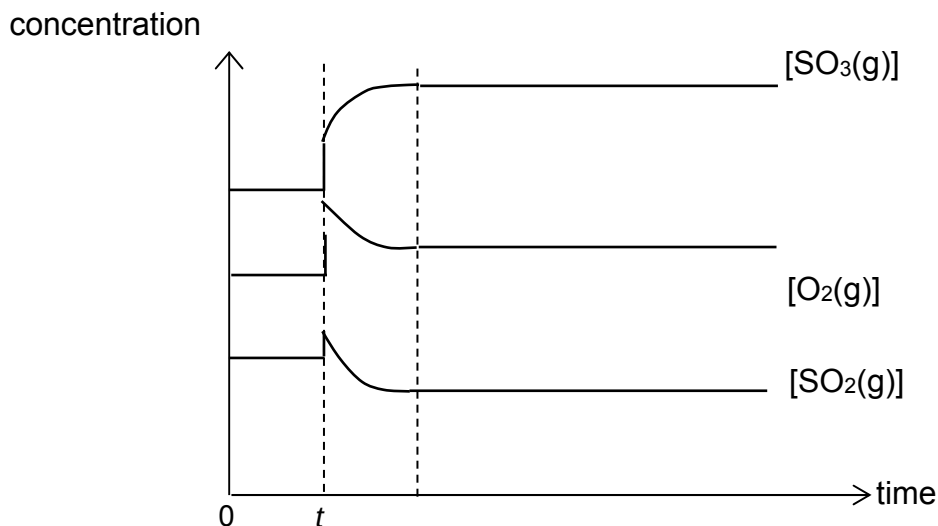
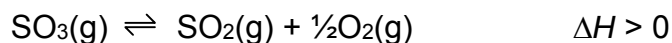
The time taken for an intense blue colour to be observed is then determined.

Experiment	Volume used / cm ³			Time / s
	1.0 mol dm ⁻³ KI	0.040 mol dm ⁻³ Na ₂ S ₂ O ₈	H ₂ O	
1	10.0	5.0	25.0	170
2	15.0	5.0	20.0	113
3	15.0	10.0	15.0	56.5
4	20.0	x	y	21.3

What are the values of **x** and **y** in Experiment 4?

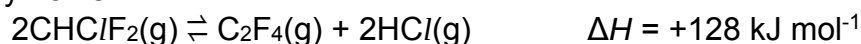
- | | | |
|----------|----------|----------|
| | x | y |
| A | 5.0 | 15.0 |
| B | 10.0 | 10.0 |
| C | 15.0 | 5.0 |
| D | 20.0 | 0.0 |

- 8 An equilibrium mixture containing gaseous SO_2 , O_2 and SO_3 is subjected to a change at time t . The mixture is allowed to reach equilibrium again. The variation of the concentrations of SO_2 , O_2 and SO_3 with time is given below.



Which of the following changes made at time t would result in the above variation of the concentrations of the gases?

- A Increasing the temperature.
 - B Decreasing the temperature.
 - C Adding a catalyst.
 - D Reducing the volume of the vessel.
- 9 Poly(tetrafluoroethene) is a polymer used as a coating in non-stick kitchen utensils and for replacement bone joints. One of the stages in the manufacture of the polymer is



Which of the following changes will increase the amount of tetrafluoroethene, C_2F_4 , present at equilibrium?

- 1 decrease pressure
- 2 increase temperature
- 3 use a catalyst

- A 1 only
- B 1 and 2 only
- C 2 and 3 only
- D 1, 2 and 3

- 10 Three unknown solutions **P**, **Q** and **R** contain a strong monobasic acid, a weak monobasic acid and a strong monoacidic base, but not necessarily in this same order. The concentration and pH of each solution are shown below.

P	Q	R
1.000 mol dm ⁻³ pH = 2.4	0.010 mol dm ⁻³ pH = 12.0	0.001 mol dm ⁻³ pH = 3.0

Which of the following statements **are** correct?

- 1 **Q** contains a strong base.
- 2 **P** contains a weak acid.
- 3 Mixing 100 cm³ of **P** and 100 cm³ of **R** produces a buffer solution.

- A 1 only is correct.
- B 1 and 2 only are correct.
- C 2 and 3 only are correct.
- D 1, 2 and 3 are correct.

- 11 The numerical values of the solubility product of AgBr, AgIO₃ and Ag₂CO₃ are given in the table below.

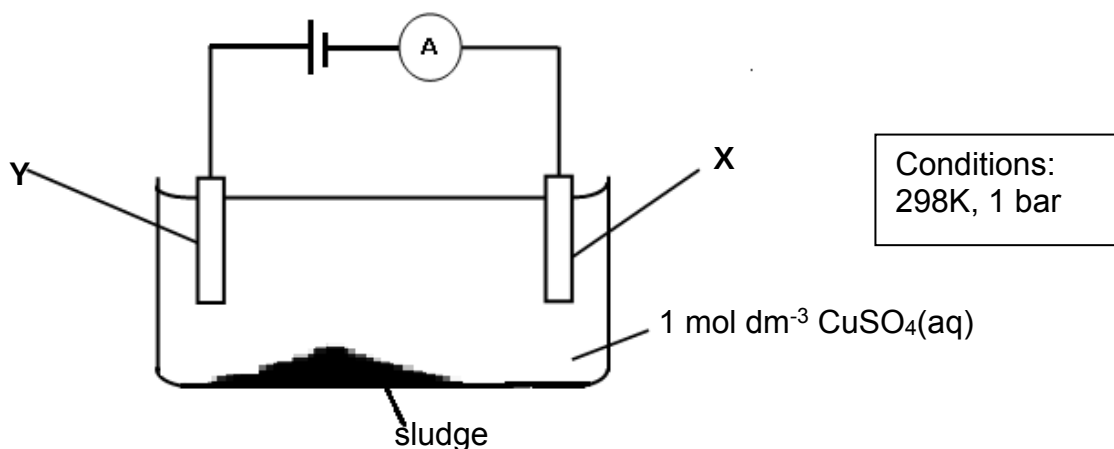
compound	solubility product
AgBr	5×10^{-13}
AgIO ₃	2×10^{-8}
Ag ₂ CO ₃	5×10^{-12}

An aqueous solution of AgNO₃ was added slowly, until in excess, to a solution containing 1.5 mol dm⁻³ of NaBr, 1.5 mol dm⁻³ of KIO₃ and 1.5 mol dm⁻³ of Na₂CO₃.

What is the correct order of precipitation of the three silver salts?

	first to precipitate	→	last to precipitate
A	AgBr	AgIO ₃	Ag ₂ CO ₃
B	AgBr	Ag ₂ CO ₃	AgIO ₃
C	Ag ₂ CO ₃	AgBr	AgIO ₃
D	AgIO ₃	Ag ₂ CO ₃	AgBr

12 Use of the Data Booklet is relevant to this question. The circuit shown in the diagram was set up to purify impure Cu, which contains Ag and Fe impurities. (In the diagram below, A represents ammeter)



What of the following represents the substance used as electrode X, the products formed at the anode and cathode, and the substance present in the sludge after some time?

	Electrode X	Products at anode	Products at cathode	sludge
A	Impure Cu	Cu^{2+}	Cu	Ag
B	Impure Cu	Fe^{2+}	Cu^{2+}	Ag
C	Pure Cu	Ag^+ , Fe^{2+}	Cu^{2+}	Zn
D	Pure Cu	Cu^{2+} , Fe^{2+}	Cu	Ag

13 Use of the Data Booklet is relevant to this question.

An electrochemical cell is set up using a $\text{Fe}^{2+}(\text{aq})|\text{Fe}(\text{s})$ half-cell and a $\text{VO}_2^+(\text{aq}), \text{VO}^{2+}(\text{aq})|\text{Pt}(\text{s})$ half-cell.

Which of the following gives a correct effect on the E_{cell} and a correct explanation for the effect when each of the changes is made to the cell separately?

Change	Effect on E_{cell}	Explanation
1 Add KCN(aq) to the $\text{Fe}^{2+}(\text{aq}) \text{Fe}(\text{s})$ half-cell	Increases	Concentration of $\text{Fe}^{2+}(\text{aq})$ decreases
2 Add water to the $\text{VO}_2^+(\text{aq}), \text{VO}^{2+}(\text{aq}) \text{Pt}(\text{s})$ half-cell	Decreases	Concentration of water increases
3 Add NaOH(aq) to $\text{Fe}^{2+}(\text{aq}) \text{Fe}(\text{s})$ half-cell	No change	pH has no effect on E_{cell}

- A 1 only
 B 1 and 3 only
 C 2 and 3 only
 D 1, 2 and 3

14 These information relates to element **S**.

- **S** is in Period 3 of the Periodic Table.
- **S** forms an acidic oxide on exposure to air.
- **S** atom has a half-filled subshell in its ground state.

What is the identity of **S**?

- A Na
- B Si
- C P
- D S

15 Coloured glass, as used in church windows, requires three oxides; one of a transition metal, and two other oxides: one macromolecular and one ionic.

Which combination is likely to produce a coloured glass?

- | | | | |
|----------|---------------------------|--------------|--------------|
| A | Al_2O_3 | MgO | SnO |
| B | P_4O_{10} | CaO | CuO |
| C | SiO_2 | CaO | CuO |
| D | SiO_2 | PbO | ZnO |

16 The carbonates of Group 2 decompose according to the following equation.



For this reaction, ΔH value increases down the group.

Carbonate	Mg	Ca	Sr	Ba
$\Delta H / \text{kJ mol}^{-1}$	+ 101	+ 178	+ 235	+ 269

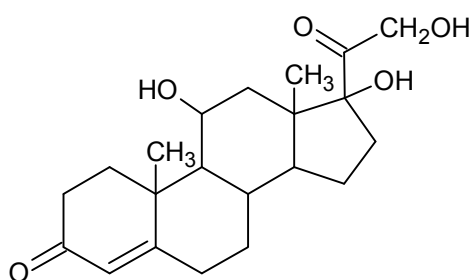
Which property best explains this trend?

- A The first and second ionisation energy of the metal
- B The lattice energy of the metal carbonate
- C The ionic radius of the metal ion
- D The proton number of the metal

- 17 A yellow-green solution is formed when concentrated hydrochloric acid is added to blue aqueous copper(II) chloride solution. Addition of potassium iodide to the mixture results in the formation of a brown solution and a white precipitate.

Which process does **not** occur in this sequence?

- A dative bond formation
 - B complex ion formation
 - C precipitation of copper(II) iodide
 - D reduction of Cu^{2+} ions
- 18 Hydrocortisone is the active ingredient in some anti-inflammatory skin cream.



hydrocortisone

How many sp^3 hybridised carbon atoms and chiral centres are present in a hydrocortisone molecule?

	sp^3 carbon atoms	chiral centres
A	16	6
B	16	7
C	17	6
D	17	7

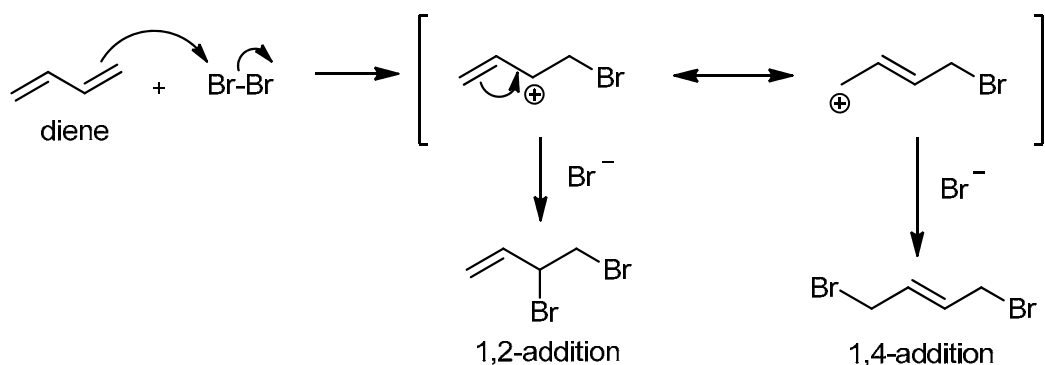
- 19 Ethane reacts with chlorine gas in the presence of ultraviolet light to form a mixture of products via free radical substitution.

Which statements about this reaction are correct?

- 1 Homolytic fission occurs only in the initiation step.
- 2 Chloroethane is formed in the propagation and termination steps.
- 3 Bond formation occurs only in the termination step.

- A 2 only
 B 2 and 3 only
 C 1 and 3 only
 D 1, 2 and 3

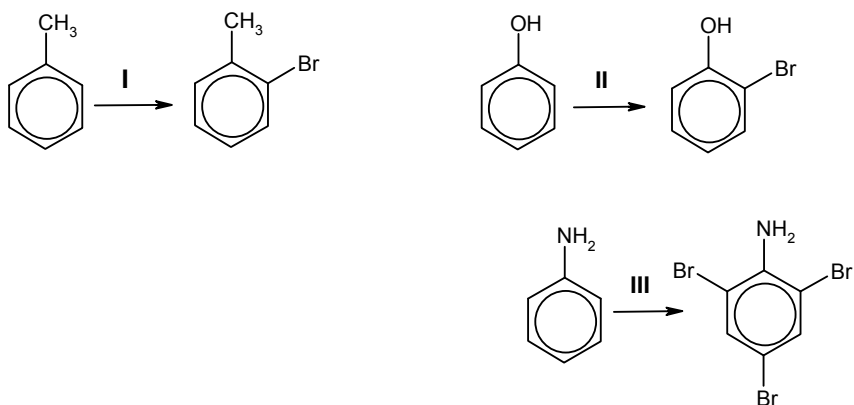
- 20 When a conjugated diene undergoes electrophilic addition with Br_2 , it forms two products through the 1,2-addition and the 1,4-addition, which is shown in the mechanism below.



Which of the following statements is **not** correct?

- A The overall rate law is second order.
 B The carbocation intermediates are resonance stabilised.
 C The 1,2-addition product formed when HCl is used is
 D The 1,4-addition product formed when ICl is used is

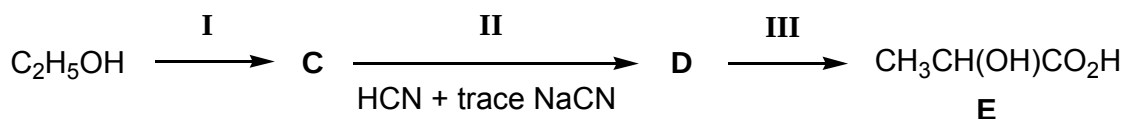
- 21 Methylbenzene, phenol and phenylamine all undergo electrophilic substitution with suitable reagents.



Which of the following could the reagents be for steps I to III?

- | | I | II | III |
|---|---------------------------------|---------------------------------|---------------------------------|
| A | Br_2, Fe | Br_2 in CCl_4 | $\text{Br}_2(\text{aq})$ |
| B | $\text{Br}_2, \text{FeBr}_3$ | $\text{Br}_2(\text{aq})$ | Br_2 in CCl_4 |
| C | $\text{Br}_2(\text{aq})$ | Br_2 in CCl_4 | Br_2, Fe |
| D | Br_2 in CCl_4 | $\text{Br}_2, \text{FeBr}_3$ | $\text{Br}_2(\text{aq})$ |
- 22 2-methylpropanoic acid can be synthesised from 1-chloropropane through a series of reactions. Which set of reagents, used in sequential order, would be the most suitable for this synthesis?
- A step I: aqueous HCN with trace KCN
step II: dilute NaOH
- B step I: ethanolic KCN
step II: dilute HCl
- C step I: aqueous KOH
step II: gaseous HCl
step III: ethanolic KCN
step IV: dilute HCl
- D step I: ethanolic KOH
step II: gaseous HBr
step III: ethanolic KCN
step IV: dilute H_2SO_4

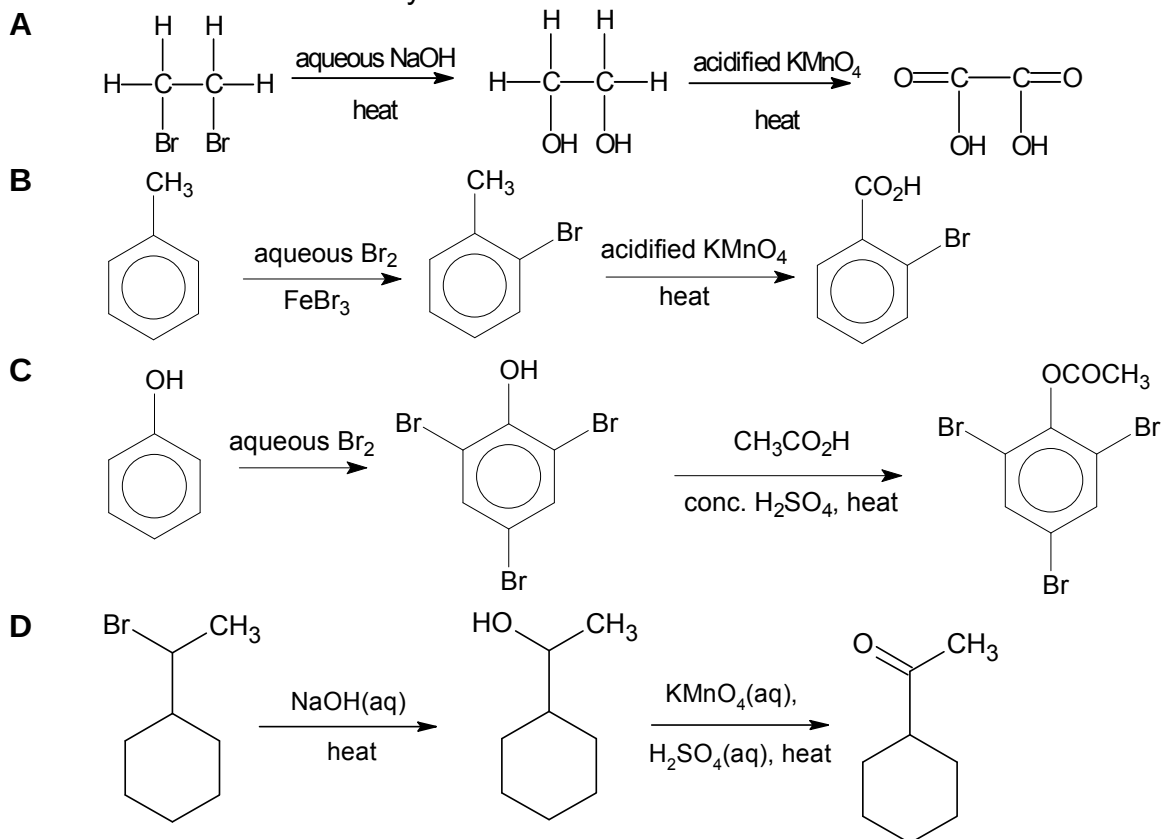
- 23 Which of the following statements on the nucleophilic substitution of 3-bromo-3-methylhexane with aqueous sodium hydroxide is **not** true?
- A The product has a chiral centre.
 B The intermediate is a carbocation.
 C The reaction produces an optically inactive mixture.
 D The OH^- nucleophile attacks 3-bromo-3-methylhexane just as the Br^- leaves.
- 24 Lactic acid (2-hydroxypropanoic acid), **E**, occurs in soured milk. It can be synthesised from ethanol by the following series of reactions.



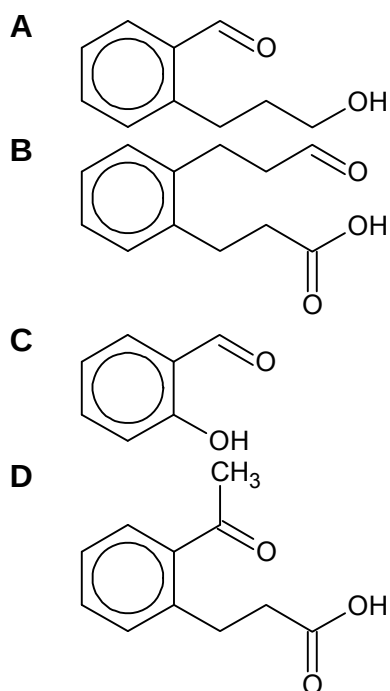
Which of the following correctly represent the reagents and conditions for step I, and the structures of compounds **C** and **D**?

	step I	C	D
A	$\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat	CH_3COOH	CH_3COCN
B	$\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, distil	CH_3CHO	$\text{CH}_3\text{CH}(\text{CN})\text{OH}$
C	$\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat	CH_3COOH	CH_3COCN
D	$\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, distil	CH_3CHO	$\text{CH}_3\text{CH}(\text{CN})\text{OH}$

25 Which is the most feasible synthesis route?



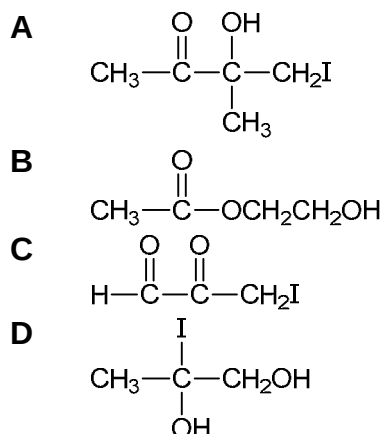
26 Compound **X** gives a positive result when treated with $[\text{Ag}(\text{NH}_3)_2]^+$ and PCl_5 but a negative result when treated with alkaline $\text{Cu}(\text{II})$ complex. What could **X** be?



27 Compound Y has the following properties.

- It changes the colour of acidified sodium dichromate(VI) from orange to green.
- It gives yellow precipitate with warm, alkaline aqueous iodine.

Which compound could be Y?

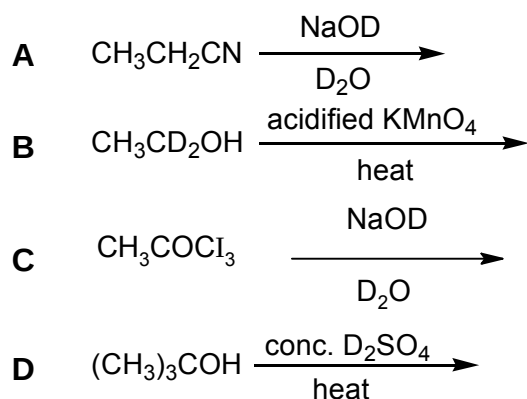


28 An ester Z was heated with aqueous NaOH and the resulting mixture was then distilled. The distillate obtained decolourised hot, acidified potassium manganate(VII), giving effervescence that forms white precipitate with aqueous $\text{Ca}(\text{OH})_2$. The residue in the distillation flask, after acidification, gave a white precipitate insoluble in water.

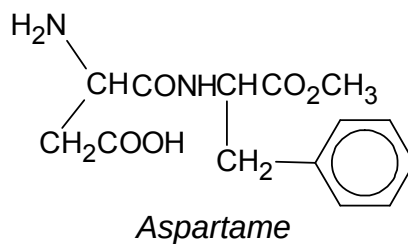
Which of these could be Z?

- A** $\text{C}_6\text{H}_5\text{CO}_2\text{CH}_3$
- B** $\text{CH}_3\text{CO}_2\text{C}_6\text{H}_5$
- C** $\text{CH}_3\text{CO}_2\text{CH}_2\text{C}_6\text{H}_5$
- D** $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{C}_6\text{H}_5$

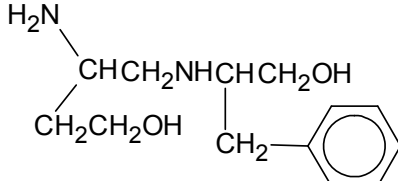
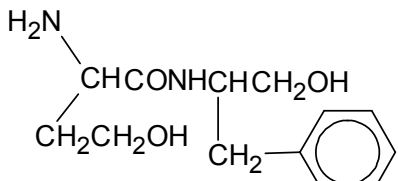
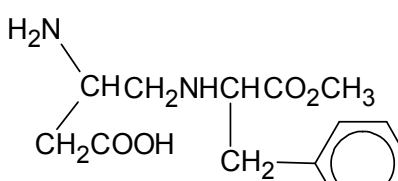
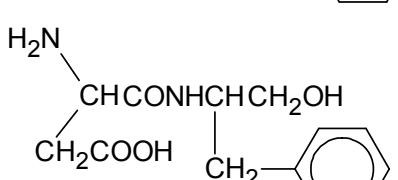
29 Which reaction produces a carbon compound incorporating deuterium, D? (D = ^2_1H , an isotope of hydrogen)



- 30 *Aspartame* is widely used as a sweetener in canned drinks.



When *Aspartame* is treated with LiAlH₄ in dry ether, what is the final product obtained?

- A** 
- B** 
- C** 
- D** 

End of Paper

Answers

1	B	11	A	21	A
2	C	12	D	22	D
3	D	13	A	23	D
4	A	14	C	24	D
5	A	15	C	25	D
6	B	16	C	26	A
7	D	17	C	27	C
8	D	18	D	28	A
9	B	19	A	29	C
10	B	20	C	30	A